Numerical Linear Algebra for Computational Science and Information Engineering

Introduction

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Numerical linear algebra: The "Why?", the "What?" and "What's next?"

Why do we need numerical linear algebra?

Prevalence of linear algebra in research and industry:

- Scientific computing (often explicitly related to problems of linear algebra)
- Simulations (e.g., discretized PDEs, ODEs, DAEs, ...)
- Machine learning (e.g., PCA, RKHS, ...)
- Optimization

Methods taught in linear algebra:

- Solving linear systems: Cramer's rule and Gaussian elimination
- Matrix factorizations: Cholesky, LU and QR
- Eigenvalue problems: Root-finding of characteristic polynomials

Limitations of methods taught in linear algebra:

- **No general exact method for eigenvalues of matrices** larger than 4x4
- **Computational infeasibility** of analytical methods **for large systems**
- **Numerical instability** leading to significant **errors**
- **Inability to handle special structures** (e.g., sparsity, implicitness)

Need for methods to solve challenging problems efficiently

What is numerical linear algebra?

What does numerical numerical linear algebra (NLA) address:

- Efficient algorithms for large-scale problems of linear algebra
- Techniques for maintaining numerical stability and accuracy
- Methods for exploiting matrix structure (e.g., sparsity, symmetry, low-rank)
- Matrix-free formulations for implicit operators
- Error analysis and conditioning

Scope of NLA:

- Focuses on both theoretical analysis (e.g., convergence, conditioning and stability) and practical implementation.
- Incorporates aspects of performance and computer architecture for method development, e.g.,
  - Repurposing solvers into preconditioners for higher performance.
  - Designing sparse data structures and BLAS kernels to optimize memory usage.

Key branches of NLA:

- **Direct methods** vs **iterative methods** (stationary vs Krylov)
- **Dense matrices** vs **sparse matrices** and **matrix-free operators**

What's coming next in numerical linear algebra?

Recent developments in NLA research show promise for significant performance improvements, but face challenges in widespread adoption:

- **Randomization**: Randomly reduces problem dimensions while preserving inherent structure with high probability. Associated challenges are
  - Ensuring convergence and assessing stability.
  - Defining and implementing reliably amortized sketching procedures.
- **Communication avoidance**: Redesigns algorithms to minimize data movement between processors or memory hierarchies. Challenges are
  - Maintaining numerical stability with reduced communication.
- **Mixed precision**: Utilizes different numerical precisions within a single algorithm to optimize performance. Associated challenges are
  - Ensuring overall stability and accuracy with lower precision components.
  - Developing adaptive strategies for precision selection.
- **Extension of state-of-the art methods to tensor data**: Assessing convergence and stability when using low-rank approximation of tensor data, e.g., Tucker and Tensor-Train formats.

Computational Mathematics @ TUM

Computational Mathematics @ TUM, Campus Heilbronn

- Chair of COmputational MAthematics @ TUM: Prof. **Hartwig Anzt**
  - Professor @ TUM since 01/2024
  - Director of the Innovative Computing Lab @ University of Tennessee (2022-23)
  - Junior Professor @ KIT (2021-22)
- Recent and ongoing PhD theses directed by Hartwig:
  - Symbolic LU factorizations on GPUs (**Tobias Ribizel**, ongoing)
  - Scalable domain decomposition on GPUs (**Fritz Göbel**, ongoing)
  - Numerical compression in scientific computing (**Thomas Grützmacher**, 2026)
  - Mixed precision algebraic multigrids on GPUs (**Yu-Hsiang Mike Tsai**, 2024)
  - Asynchronous and batched iterative solvers on GPUs (**Pratik Nayak**, 2023)
- Ongoing projects in COMA:
  - **Ginkgo**: Portable high-performance numerical linear algebra library
  - **Sparse BLAS**: Basic linear algebra subprograms for sparse matrices
  - **ICON**: Large legacy codebase for climate modeling
  - **OGL/OpenFOAM**: Large legacy codebase for computational fluid dynamics
  - **MicroCARD**: Numerical modeling of cardiac electrophysiology
  - **nekRS**: Fast and scaleable computational fluid dynamics software package

Computational Mathematics @ TUM, Campus Heilbronn, cont'd$_1$

- Course Instructor:



Nicolas Venkovic, Postdoc. in COMA @ TUM
  - Postdoctoral Researcher @ TUM since 07/2024
  - PhD in Applied Mathematics & Scientific Computing @ Cerfacs (09/2023)
  - MSE in Applied Mathematics & Statistics @ Johns Hopkins University (2018)
- Recent and ongoing research efforts (see venkovic.github.io/research):
  - **Dense Convolution Kernels**
  - **Locally Optimal Iterative Methods for**:
    - **Strucutred Matrix Inverses**, **Eigenpairs**, **Low-Rank** and **Non-Negative Factorizations**
  - **Applications**:
    - **Stochastic Preconditioning**, **Matrix Recovery**, **Parallelization**, ...

Computational Mathematics @ TUM, Campus Heilbronn, cont'd$_2$

- Teaching Assistant:



Amir Bouslama, Master student @ TUM
  - Master student in Information Engineering @ TUM since Winter 2025/26
  - BSc in Information Engineering @ TUM (Summer 2025)

- Responsibilities:
  - **Exercise** (Problem Sets) + **Practice** (Julia) Sessions
    1 Session per 3 Meetings (see schedule on Moodle)
  - **Grading**
- Research interests:
  - **Numerical Methods** and **Simulation Software**

Class overview

Objectives

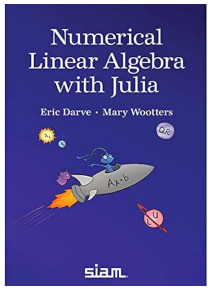
Upon completion of the class, students should:

- Understand why numerical linear algebra (NLA) is essential in practice, and what it encompasses.
- Be familiar with the challenges of the main problems and procedures of NLA, such as orthogonalization, least-squares solving, factorization, linear solving, preconditioning, eigenvalue solving, ...
- Understand the definitions, implementations and main properties of the methods presented in class for solving these problems.
- Recognize some pathological behaviors of these methods and be able to explain unwanted numerical behaviors.
- Know the strengths and limits of the presented methods, and know when to use them in practice.
- Have the foundational background to explore methods for solving NLA problems beyond those covered in class.

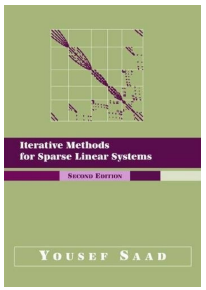
Content

- We will cover the following core topics:
  - `Lecture 01`: Essentials of linear algebra
  - `Lecture 02`: Floating-point arithmetic and error analysis
  - `Lecture 03`: Direct methods for dense linear systems
  - `Lecture 04`: Sparse data structures and basic linear algebra subprograms
  - `Lecture 05`: Introduction to direct methods for sparse linear systems
  - `Lecture 06`: Orthogonalization and least-squares problems
  - `Lecture 07`: Basic iterative methods for linear systems
  - `Lecture 08`: Basic iterative methods for eigenvalue problems
  - `Lecture 09`: Locally optimal block preconditioned conjugate gradient
  - `Lecture 10`: Arnoldi and Lanczos procedures
  - `Lecture 11`: Jacobi-Davidson methods
  - `Lecture 12`: Krylov subspace methods for linear systems
  - `Lecture 13`: Preconditioned iterative methods for linear systems
- All lectures will begin by establishing formal foundations. Most lectures will be followed by practice sessions (with Amir, see schedule on Moodle) using **Julia notebooks** and/or **problem sets** to test, visualize, and deepen your understanding of the methods and concepts introduced.

Reading material

- **Main references**:
  - **Lecture slides**, **Julia notebooks** and **problem sets** uploaded gradually on Moodle, throughout the semester.
    Content also uploaded at venkovic.github.io/NLA-for-CS-and-IE.
  - 

    Darve, E., & Wootters, M. (2021). Numerical Linear Algebra with Julia. Society for Industrial and Applied Mathematics (SIAM).†
  - GitHub repository:
    `https://github.com/EricDarve/numerical_linear_algebra`

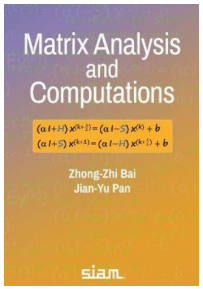
- **Most used supplemental reference #1**:
  - 

    Saad, Y. (2003). Iterative Methods for Sparse Linear Systems. Society for Industrial and Applied Mathematics (SIAM).†

† SIAM members get 30% off book prices. SIAM membership is free for students.

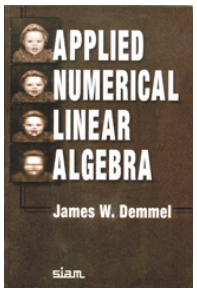
Reading material, cont'd$_1$

- **Most used supplemental reference #2**

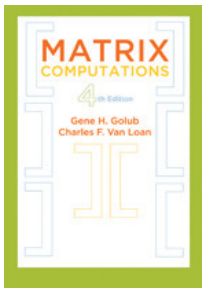


- Bai, Z. Z., & Pan, J.-Y. (2021). Matrix Analysis and Computations. Society for Industrial and Applied Mathematics (SIAM).†

- **Other useful supplemental references**:



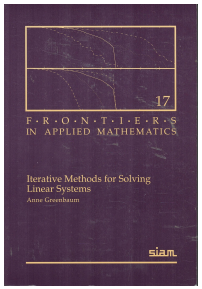
- Demmel, J. W. (1997). Applied Numerical Linear Algebra. Society for Industrial and Applied Mathematics (SIAM).†



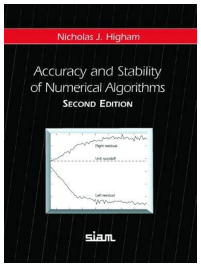
- Golub, G. H., & Van Loan, C. F. (2013). Matrix Computations. JHU press.

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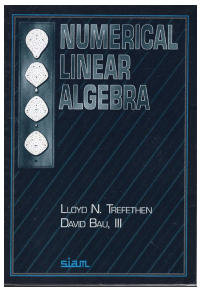
Reading material, cont'd$_2$



- Greenbaum, A. (1997). Iterative Methods for Solving Linear Systems. Society for Industrial and Applied Mathematics (SIAM).†



- Higham, N. J. (2002). Accuracy and Stability of Numerical Algorithms. Society for Industrial and Applied Mathematics (SIAM).†



- Trefethen, L. N., & Bau, D. (2022). Numerical Linear Algebra. Society for Industrial and Applied Mathematics (SIAM).†

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Course evaluation

- **Final exam (determines base grade)**:

  1.5 hr-long written exam @ Campus Heilbronn

  - Questions in relation to the problem sets, bonus tests, material presented in class, and practice sessions.
- **Bonus tests policy**:
  - 4 in-class closed-book bonus tests at set dates (see schedule on Moodle).
  - Deadline: 2 weeks after the end of the lecture.
  - Bonus points up to one step on final grade.
    - Example: 2.0 becomes 1.7.
    - Notes: This upgrade applies to passing grades only.
      1.0 remains the highest possible grade.